Role 3 — Comparative Statics for the FCL / Bitcoin-Mining Carbon Model: Math, Proofs, and Numerical Results

Role 3

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1 Model recap

Residual demand is

$$x \;=\; D - K_R\,\eta, \qquad D \in [0, \bar{D}],\ \eta \in [0,1],$$

with CDF $F(\,\cdot\,; K_R)$. The utility chooses capacities (K_F, K_R, K_M) and dispatches the flexible computing load by the threshold rule $d(x) = \min\{(K_F - x)_+, K_M\}$ (the optimal within-period contract; Role 1). We treat the per-MWh computing margin R as exogenous, consistent with Role 1's baseline; the miner-side market structure (oligopoly demand, mining price) is a B-workstream / miner-side object and is *outside A1's scope*, so it does not enter the comparative statics below.

Assumption 1 (interior regime; identical to Role 1/Role 4).

$$R > c + C_M, \qquad R < c + C_F + C_M, \qquad \beta + p > c + C_F, \qquad \beta + p > R.$$

2 No-FCL benchmark

Without FCL the planner solves

$$\max_{K_F, K_R}\; p\,\mathbb{E}[\bar{D} - (x - K_F)_+] - c\,\mathbb{E}[\min(x_+, K_F)] - (\beta + p)\mathbb{E}[(x - K_F)_+] - C_F K_F - C_R K_R.$$

First-order conditions give

$$F(K_F^\circ; K_R^\circ) = 1 - \frac{C_F}{\beta + p - c}, \tag{1}$$

$$\mathbb{E}[\eta \mid x > 0]\big(c\,F(0) + (\beta + p)(1 - F(K_F^\circ))\big) = C_R. \tag{2}$$

3 Joint optimum with FCL

With FCL the FCL dispatch is $d(x) = \min\{(K_F - x)_+, K_M\}$. The planner's FOCs (eqs. 12,13,15 of the paper) are

$$F(K_F^\star; K_R^\star) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \tag{3}$$

$$F(K_F^\star - K_M^\star; K_R^\star) = \frac{C_M}{R - c}, \tag{4}$$

together with a renewable FOC of the same form as (**??**) but with the FCL dispatch added in the inner region.

4 Proposition 1 (distribution-free)

Proposition 1. *Under Assumption 1 and $C_M < \bar{C}_M$ with*

$$\bar{C}_M \;=\; (R-c)\,\frac{\beta+p-c-C_F}{\beta+p-c},$$

introducing FCL strictly increases the utility's profit and strictly decreases blackout probability, for any continuous residual-demand distribution F.

Proof. Profit. The FCL planner's program nests the no-FCL one ($K_M = 0$ is feasible). Optimality therefore weakly raises profit; the strict inequality follows because at $K_M = 0$ the gradient of the Lagrangian in K_M equals $(R-c)\big[1-F(K_F^\circ)\big]-C_M$, which is positive iff $C_M < \bar{C}_M$, so the optimum has $K_M^\star > 0$ and profit strictly rises.

Blackout. Comparing (**??**) and (**??**): the right-hand side of (**??**) exceeds that of (**??**) in the interior regime, so the target quantile is higher and $K_F^\star > K_F^\circ$ when K_R is held fixed. Lemma **??** only reinforces this through $K_R^\star \geq K_R^\circ$. Hence $\Pr(x > K_F^\star) < \Pr(x > K_F^\circ)$. □

Lemma 1 (Capacity substitution)**.** *In the interior regime, $K_R^\star \geq K_R^\circ$, with strict inequality whenever $K_M^\star > 0$.*

Proof. Comparing renewable FOCs: the FCL version adds a strictly positive term $(R-c)\mathbb{E}[\eta\,\mathbf{1}\{0 < x < K_F - K_M\}]$ to the marginal benefit of renewable capacity, while the blackout term is (weakly) smaller because $K_F^\star \geq K_F^\circ$. Net marginal benefit of K_R rises, so the optimal K_R rises. □

5 Carbon decomposition (eq. 17) — two forces, matching Role 1

We adopt Role 1's decomposition verbatim to avoid any ambiguity. The *reduction* in carbon from introducing FCL (no-FCL emissions minus with-FCL emissions) splits into two opposing forces:

$$\underbrace{c\,\mathbb{E}[\text{fossil}^\circ] - c\,\mathbb{E}[\text{fossil}^\star]}_{\Delta\text{Carbon (reduction)}} \;=\; \underbrace{\text{capacity-substitution effect}}_{\geq 0 \text{ when } K_R^\star > K_R^\circ \ (\leq 0 \text{ otherwise})} \;-\; \underbrace{\text{utilization effect}}_{>0,\ \text{distribution-free}}. \tag{5}$$

The **utilization effect** is Role 1's Term (II): running computing on spare fossil strictly *raises* fossil output in every state, so it is always positive and reduces the saving. The **capacity-substitution effect** is Role 1's Term (I): the larger renewable fleet $K_R^\star > K_R^\circ$ (Lemma **??**) shifts residual demand left and *lowers* fossil output. Carbon falls iff the second beats the first — exactly Role 1's condition (C). Our numerics below confirm both signs and map where the net is negative.

6 Section 5.3 (Case C, short run)

Corollary 1. *If $(K_F^{(0)}, K_R^{(0)})$ are exogenously fixed and only K_M is optimised, then the carbon reduction $\Delta Carbon \leq 0$ — FCL weakly* increases *emissions (Role 1, §5.3 / eq. 30). Equality holds iff $F(K_F^{(0)}; K_R^{(0)}) \leq C_M/(R-c)$, i.e. $K_M^\star = 0$.*

Proof. Capacities are frozen, so the capacity-substitution effect in (**??**) vanishes identically. Only the positive utilization effect remains, so the reduction is ≤ 0. □

Quantity	Below $\bar{C}_M$ (16 sc.)	All (40 sc.)
Profit improves with FCL	16/16	16/40
Blackout probability drops	16/16	16/40
Fossil utilisation rises	–	30/40
Carbon drops	–	10/40

Table 1: Bullets 1–2: Proposition **??** holds in every scenario with $C_M < \bar{C}_M$.

Quantity	Scenarios
Carbon weakly rises (Case C, capacities fixed)	96/96
$K_R^\star > K_R^\circ$ (capacity substitution)	32/32
Carbon strictly falls (general model)	2/32

Table 2: Bullets 3–4: Corollary **??** confirmed; carbon reduction concentrates in right-skewed F with low C_R.

7 Numerical confirmation (40 000-sample Monte Carlo)

Distributions tested. Uniform; Beta$(0.5, 3)$, Beta$(2, 5)$ (right-skewed); Beta$(2, 2)$ (symmetric); Beta$(5, 2)$ (left-skewed); lognormal(σ=0.6); exponential(0.3); Pareto(2.5).

8 Sync notes for Role 1

- Profit and blackout halves of Proposition **??** verified distribution-free across 8 families whenever $C_M < \bar{C}_M$.
- Utilization effect (Role 1 Term (II)) is positive — it *raises* fossil output — in every scenario, matching the distribution-free sign.
- Capacity-substitution channel $K_R^\star > K_R^\circ$ holds in 32/32 scenarios — this is the binding driver of the carbon sign.
- Left-skewed F with high C_R reverses the carbon sign; this is the boundary region for Role 1's general statement.